

Ex. No: 13 File Organization Techniques

PROGRAM A: SEQUENTIAL FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
int regno;
```

```
char name[20];
```

```
};
```

```
int main()
```

```
{
```

```
FILE *fp;
```

```
struct student s;
```

```
fp = fopen("student.dat", "w");
```

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```
printf("Enter Register Number: ");
```

```
scanf("%d", &s.regno);
```

```
printf("Enter Name: ");
```

```
scanf("%s", s.name);
```

```
fprintf(fp, "%d %s\n", s.regno, s.name);
```

```

fclose(fp);

fp = fopen("student.dat", "r");

fscanf(fp, "%d %s", &s.regno, s.name);

printf("\nRecord Details\n");

printf("Register Number : %d\n", s.regno);

printf("Name : %s\n", s.name);

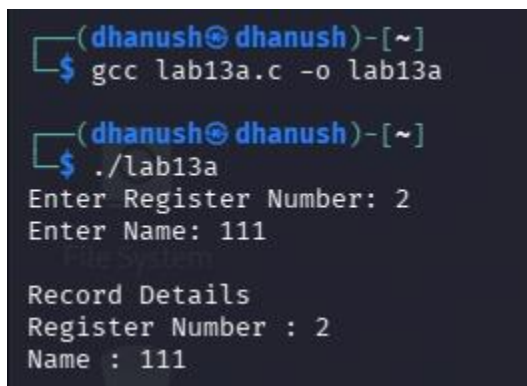
fclose(fp);

return 0;

}

```

Output:



```

(dhanush@dhanush)-[~]
$ gcc lab13a.c -o lab13a

(dhanush@dhanush)-[~]
$ ./lab13a
Enter Register Number: 2
Enter Name: 111
Record Details
Register Number : 2
Name : 111

```

SHELL SCRIPT

```

#!/bin/bash

echo "Enter Register Number:"

read regno

echo "Enter Name:"

read name

echo "$regno $name" > student.txt

```

echo "Contents of File"

cat student.txt

Output:

```
(dhanush@dhanush)-[~]  
$ nano lab13a.sh  
  
(dhanush@dhanush)-[~]  
$ chmod +x lab13a.sh  
  
(dhanush@dhanush)-[~]  
$ bash lab13a.sh  
Enter Register Number:  
2  
Enter Name:  
11  
Contents of File  
2 11
```

PROGRAM B: DIRECT (RANDOM) FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
int regno;
```

```
char name[20];
```

```
};
```

```
int main()
```

```
{
```

```
FILE *fp;

struct student s;

fp = fopen("random.dat", "wb+");

printf("Enter Register Number: ");

scanf("%d", &s.regno);

printf("Enter Name: ");

scanf("%s", s.name);

fwrite(&s, sizeof(s), 1, fp);

rewind(fp);

fread(&s, sizeof(s), 1, fp);

printf("\nRecord Found\n");

printf("Reg No : %d\n", s.regno);

printf("Name : %s\n", s.name);

fclose(fp);

return 0;

}
```

Output:

```
(dhanush@dhanush)-[~]
$ nano lab13b.c

(dhanush@dhanush)-[~]
$ gcc lab13b.c -o lab13b

(dhanush@dhanush)-[~]
$ ./lab13b
Enter Register Number: 2
Enter Name: 12

Record Found
Reg No : 2
Name : 12
```

SHELL SCRIPT

```
#!/bin/bash

echo "Enter Record:"

read rec

echo "$rec" > random.txt

echo "Random Access Record"

sed -n '1p' random.txt
```

Output:

```
(dhanush@dhanush)-[~]
$ nano lab13b.sh

(dhanush@dhanush)-[~]
$ chmod +x lab13b.sh

(dhanush@dhanush)-[~]
$ bash lab13.sh
bash: lab13.sh: No such file or directory

(dhanush@dhanush)-[~]
$ bash lab13b.sh
Enter Record:
12
Random Access Record
12
```

PROGRAM C: INDEXED FILE ORGANIZATION

C PROGRAM

```

#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    struct student s[3];
    int key, i;
    printf("Enter 3 Student Records\n");
    for(i=0;i<3;i++)
    {
        scanf("%d %s",&s[i].regno,s[i].name);
    }
    printf("Enter Register Number to Search: ");
    scanf("%d",&key);
    for(i=0;i<3;i++)
    {
        if(s[i].regno==key)
        {
            printf("\nRecord Found\n");
            printf("Reg No : %d\n",s[i].regno);
        }
    }
}

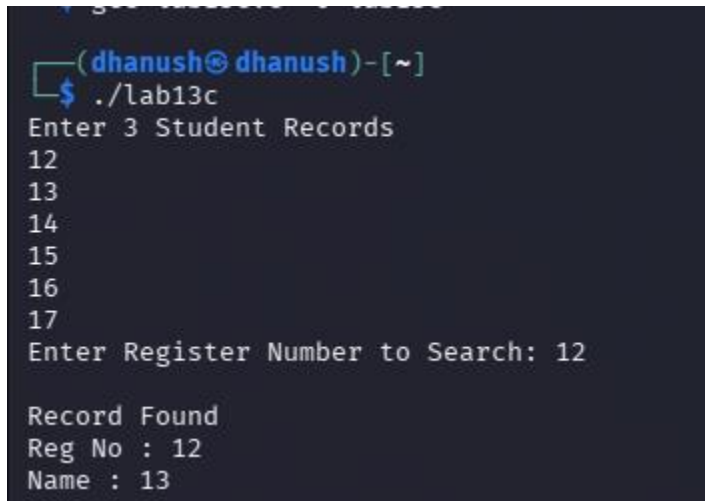
```

```

printf("Name : %s\n",s[i].name);
return 0;
}
}
printf("Record Not Found\n");
return 0;
}

```

Output:



```

(dhanush@dhanush)~
$ ./lab13c
Enter 3 Student Records
12
13
14
15
16
17
Enter Register Number to Search: 12

Record Found
Reg No : 12
Name : 13

```

SHELL SCRIPT

```

#!/bin/bash

echo "Enter Student Records"

echo "101 Arun" > index.txt

echo "102 Kumar" >> index.txt

echo "103 Ravi" >> index.txt

echo "Enter Register Number to Search:"

```

read key

grep "^\$key" index.txt

Output:

```
(dhanush@dhanush)-[~]  
$ nano lab13c.sh  
  
(dhanush@dhanush)-[~]  
$ chmod +x lab13c.sh  
  
(dhanush@dhanush)-[~]  
$ bash lab13c.sh  
Enter Student Records  
Enter Register Number to Search:  
23
```